Set 3 4 5, Excersise 6

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Bounds for European Put Options (3rd Relation)

Theorem: The price of a European put option p must satisfy the following bounds:

$$\max(PV(X) - S_0, 0) \leq p \leq PV(X) \tag{1}$$

where:

- p: Current price of the European put option
- S_0: Current price of the underlying stock
- X: Strike price
- $PV(X)$: Present Value of the strike price (typically Xe^{-rT})
- S_T: Stock price at maturity T

Part A: The Upper Bound ($p \leq PV(X)$)

Strategy 1: Buy the Put option. Cost $= p$.
Strategy 2: Keep cash equivalent to $PV(X)$. Cost $= PV(X)$.
At maturity T:

- The payoff of the Put is $\max(X - S_T, 0)$. The maximum possible value is X (which occurs if the stock price drops to 0).
- The cash in Strategy 2 grows to exactly X.

Since the payout of Strategy 1 can never exceed the sure payout of Strategy 2, the current price of Strategy 1 cannot be higher than Strategy 2, so:

$$p \leq PV(X) \tag{2}$$

Part B: The Lower Bound ($p \geq PV(X) - S_0$)

Since option prices are non-negative ($p \geq 0$), we only need to prove that $p + S_0 \geq PV(X)$.
Portfolio A: Buy one Put option (p) and one share of stock (S_0). Cost $= p + S_0$.
Portfolio B: Keep cash equivalent to $PV(X)$. Cost $= PV(X)$.
At maturity T:

- **Payoff of A:**

$$\text{Payoff}_A = \begin{cases} (X - S_T) + S_T = X & \text{if } S_T < X \text{ (Put exercised)} \\ 0 + S_T = S_T & \text{if } S_T \geq X \text{ (Put worthless)} \end{cases}$$

Therefore, $\text{Payoff}_A = \max(X, S_T)$.

- **Payoff of B:** The cash grows to exactly X.

Since $\max(X, S_T) \geq X$ in all scenarios, Portfolio A is always worth at least as much as Portfolio B. To avoid arbitrage, its initial cost must be at least as high:

$$\begin{aligned} p + S_0 &\geq PV(X) \\ p &\geq PV(X) - S_0 \end{aligned} \tag{3}$$

Conclusion: Combining the non-negativity of options, Part A, and Part B:

$$\boxed{\max(PV(X) - S_0, 0) \leq p \leq PV(X)}$$

European Call-Put Parity (5th Relation)

Theorem: For European options with the same strike X and maturity T:

$$c - p = S_0 - PV(X) \tag{4}$$

Arbitrage Argument

We construct two portfolios at time $t = 0$:

Portfolio A (Fiduciary Call):

- Buy one European Call option (c)
- Invest cash equal to the Present Value of the strike ($PV(X)$)

$$\text{Initial Cost}_A = c + PV(X)$$

Portfolio B (Protective Put):

- Buy one European Put option (p)
- Buy one share of the underlying stock (S_0)

$$\text{Initial Cost}_B = p + S_0$$

Payoff Analysis at Maturity

We examine the value of both portfolios at time T depending on the spot price S_T:

Scenario	Portfolio A Payoff	Portfolio B Payoff
$S_T \leq X$	Call expires (0) + Cash (X) $= \mathbf{X}$	Put exercised ($X - S_T$) + Stock (S_T) $= \mathbf{X}$
$S_T > X$	Call exercised ($S_T - X$) + Cash (X) $= \mathbf{S_T}$	Put expires (0) + Stock (S_T) $= \mathbf{S_T}$

Conclusion

Both portfolios provide the exact same payoff profile: $\max(S_T, X)$. According to the **Law of One Price** (Static Arbitrage), if two portfolios have identical future payoffs in all states of the world, they must have the same current price.

Therefore:

$$\begin{aligned} \text{Cost}_A &= \text{Cost}_B \\ c + PV(X) &= p + S_0 \\ c - p &= S_0 - PV(X) \end{aligned} \tag{5}$$

$$\boxed{c - p = S_0 - PV(X)}$$